

■ 3.4.3 Simultaneous Equations

You can give `Solve` a list of simultaneous equations to solve. `Solve` can find explicit solutions for a large class of simultaneous polynomial equations.

Here is a simple linear equation with two unknowns.

```
In[1]:= Solve[ { a x + b y == 1, x - y == 2 } ,
               {x, y} ]
```

```
Out[1]= {{x -> -(1/(-a - b)) - (2 b)/(-a - b),
          y -> -(1/(-a - b)) + (2 a)/(-a - b)}}
```

Here is a more complicated example. The result is a list of solutions, with each solution consisting of a list of transformation rules for the variables.

```
In[2]:= Solve[{x^2 + y^2 == 1, x + y == a}, {x, y}]
```

```
Out[2]= {{x -> (2 a - Sqrt[4 a^2 - 8 (-1 + a^2)]) / 4,
          y -> (2 a + Sqrt[4 a^2 - 8 (-1 + a^2)]) / 4},
         {x -> (2 a + Sqrt[4 a^2 - 8 (-1 + a^2)]) / 4,
          y -> (2 a - Sqrt[4 a^2 - 8 (-1 + a^2)]) / 4}}
```

You can use the list of solutions with the `/.` operator.

```
In[3]:= x^3 + y^3 /. % /. a -> 0.7
Out[3]= {0.8785, 0.8785}
```

Even when `Solve` cannot find explicit solutions, it often can “unwind” simultaneous equations to produce a symbolic result in terms of nested `Roots` functions.

```
In[4]:= Solve[{x^3 + y^3 == x y, x + y + x y == 1},
               {x, y}]
```

```
Out[4]= {ToRules[Roots[4 x ==
                      3 - 3 y + y^2 - 2 y^3 - 2 y^4 - y^5, x,
                      Using ->
                      Roots[-4 y + 2 y^2 + y^3 + 4 y^4 + 3 y^5 + y^6
                      == -1, y]]]}
```

You can then use `N` to get a numerical result.

```
In[5]:= N[ % ]
Out[5]= {{x -> -1.74543 + 1.2018 I,
          y -> -1.74543 - 1.2018 I},
          {x -> -1.74543 - 1.2018 I, y -> -1.74543 + 1.2018 I},
          {x -> -0.17169 + 1.14626 I,
          y -> -0.17169 - 1.14626 I},
          {x -> -0.17169 - 1.14626 I,
          y -> -0.17169 + 1.14626 I},
          {x -> 0.507861, y -> 0.326382},
          {x -> 0.326382, y -> 0.507861}}
```

The variables that you use in `Solve` do not need to be single symbols. Often when you set up large collections of simultaneous equations, you will want to use expressions like `a[i]` as variables.

Here is a list of three equations for the `a[i]`.

```
In[6]:= Table[ 2 a[i] + a[i-1] == a[i+1], {i, 3} ]
Out[6]= {a[0] + 2 a[1] == a[2], a[1] + 2 a[2] == a[3],
          a[2] + 2 a[3] == a[4]}
```

This solves for some of the `a[i]`.

```
In[7]:= Solve[ % , {a[1], a[2], a[3]} ]
Out[7]= {{a[1] ->  $-\frac{5 a[0]}{12} + \frac{a[4]}{12}$ , a[2] ->  $\frac{a[0]}{6} + \frac{a[4]}{6}$ ,
          a[3] ->  $\frac{-a[0]}{12} + \frac{5 a[4]}{12}$ }}
```

<code>Solve[eqns, {x₁, x₂, ...}]</code>	solve <i>eqns</i> for the specific objects <i>x_i</i>
<code>Solve[eqns]</code>	try to solve <i>eqns</i> for all the objects that appear in them

Solving simultaneous equations.

If you do not explicitly specify objects to solve for, `Solve` will try to solve for all the variables.

```
In[8]:= Solve[ { x + y == 1, x - 3 y == 2 } ]
Out[8]= {{x ->  $\frac{5}{4}$ , y ->  $-\frac{1}{4}$ }}
```

```
Solve[{lhs1==rhs1, lhs2==rhs2, ...}, vars]
Solve[lhs1==rhs1 && lhs2==rhs2 && ..., vars]
Solve[{lhs1, lhs2, ...} == {rhs1, rhs2, ...}, vars]
```

Ways to present simultaneous equations to `Solve`.

If you construct simultaneous equations from matrices, you typically get equations between lists of expressions.

```
In[9]:= {{3,1},{2,-5}}.{x,y}=={7,8}
Out[9]= {3 x + y, 2 x - 5 y} == {7, 8}
```

`Solve` converts equations involving lists to lists of equations.

```
In[10]:= Solve[%, {x, y}]
Out[10]= {{x ->  $\frac{43}{17}$ , y ->  $-\left(\frac{10}{17}\right)}$ }
```

You can use `LogicalExpand` to do the conversion explicitly.

```
In[11]:= LogicalExpand[%%]
Out[11]= 3 x + y == 7 && 2 x - 5 y == 8
```